VU Medical Zone

BT 504 Current Final terms Fall 2020

M.C.Q

1. Actin, Coronin, Dystrophin, Keratin and Tubulin are which type of protein_____. **Cytoskeleton Protein**

2. _____is the field in which computational and statistical techniques are applied to derive biological information from genome sequences. **Genome informatics**

3. _____ genes have putative orthologs in the worm. **30%**

4. In population genomic less than_____ of human genome is coding sequence. **2%**

5. Most plants and animals have genomes greater than_____. **100 Mb**

6. Many fungal viruses contain _____ genomes. **dsRNA**

7. Yeast nuclear genome has _____ chromosomes. **16**

8. Similarity between genomes of Man – Yeast_____. **23%**

9. The number of physical maps in human_____. 24

10. An entire genome or single chromosome may be represented in a _____clone library. **YAC/BAC**

11. In human the nucleotide sequence of the estimated_____ bp. **3 billion**

12. In HPG the sub-clone Library Construction the DNA fragmented by_____. **Sonication or RE cut**

13. Transcribed in the precursor mRNA sequence but is ultimately removed by RNA splicing is called_____. **Introns**

14. _____are sites of storage and/or degradation of mRNA. **Processing bodies**

15. Xist appears to recruit a specific _____ isoform which maintains the chromosome in inactive state. **Histone**

16. In humans, Dicer is encoded by the _____ gene. **DICER1**

17. In sequencing technique purines are depurinated by _____. **Formic acid**

18. Which one of the following is used in Clone-by-clone shotgun sequencing? **YAC/BAC**

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19. luciferase and _____ generate light by conversion of luciferin to oxyluciferin. **ATP**

20. The accuracy of Oxford Nanopore is _____. **75%**

21. A tangled knot One Gene means _____. **One Polypeptide**

22. The portion of a gene's DNA or RNA, composed of exons, that codes for protein_____. **Coding DNA sequence**

23. GO stands for_____. **Gene Ontology**

24. Like TSS, the flanks of splice junctions show ____ distributions of nucleotides in certain positions. **Biased**

25. In Quality screening, screening is done while satisfying the following matrices except? **ORF**

26. De Burjn approach for genome assembly is most widely applied to short reads from the Solexa and _____Platforms. **SOLiD**

27. Which one of the following is not a problem in graph construction? **Quality control**

28. Less than _____of human diseases result from a single gene defect. **2%**

29. Secondary structure of proteins refers to the ____ conformation of some part of a polypeptide. **Local**

30. Interacting segments of polypeptide chains are held in their characteristic tertiary positions by different kinds of_____. **Weak interactions**

31. A multimer with just a few subunits is called as_____. **Oligomer**

32. 2D-gels allow the visualization of _____ proteins, whereas _____ proteins frequently cannot be detected. **abundant ,less abundant**

33. The diversity of stationary phases and separation modes gives HPLC considerable_____. **Resolving power**

34. Proteins concentration (and activity) may be different than gene expression due to _____ modification. **Post-translational**

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Subjective

Qno1: Write down the function of miRNA in Tumor Suppressors and Oncogenes. 3 marks

• miRNAs can function as tumour suppressors and oncogenes.

• Gene therapies that use miRNAs might be an effective approach to blocking tumour progression.

• miRNAs such as let-7, which has been shown to negatively regulate the Ras oncogenes, and miR- 15 and miR-16, which negatively regulate BCL2, are promising candidates for cancer treatment.

Qno2: Write down the mechanism of SMART. 5 marks

• **Single Molecule Real Time Sequencing**

• Utilizes the zero-mode waveguide (ZMW), developed in the laboratories of Harold G. Craighead and Watt W. Webb at Cornell University.

• ZMW guides light energy into a volume that is small in all dimensions compared to the wavelength of the light.

• Nanophotonic structure, a circular hole in an aluminum film on a silica substrate.

• With an active polymerase immobilized at the bottom of each ZMW, nucleotides diffuse into the ZMW chamber.

• A, C, G and T are labeled with a different fluorescent dye having a distinct emission spectrum.

• Nucleotides held by the polymerase prior to incorporation emit an extended signal that identifies the base being incorporated.

Qno3: Differentiate between ORF and CDS. 2 marks

It is the part of gene that has a potential to code for proteins comes in triples called codons, beginning and ending with a unique translation start (ATG) and one of three stop (TAA, TAG, TGA) codons. ORF is usually predicted based on DNA sequence and not proven to be transcribed.

The portion of a gene's DNA or RNA, composed of exons, that codes for protein. CDS is transcribed and is coding for something.

Qno4: Differentiate between Contig and Scaffold. 2 marks

Contig: A consensus sequence of DNA that has been assembled from overlapping DNA fragments.

Scaffolds: One or more contigs linked together by unknown sequence.

Qno5: What are Proteins Structure Determination Techniques? 3 marks

• X-ray crystallography reliable but slow, not all protein crystallize.

• Computer structureprediction programs not reliable for all proteins.

Qno6: What is peptide bond? 3 marks

In a protein molecule, each amino acid residue is joined to its neighbor by a specific type of covalent bond which is called **Peptide Bond**.

Qno7: Why Electrophoresis is used for the separation of Protein fragment? 5 marks

• Capillary electrophoresis (CE) operates on the same general principal as IEF.
• Proteins placed in an electric field will migrate to a point in a pH gradient where they display an overall neutral change.
• The performance of the analysis in a microcapillary tube provides greatly enhanced resolution over the preparative IEF techniques discussed earlier.
• CE offers the greatest resolution of all peptide analytical techniques and can be coupled directly to MS instruments.
• CE thus has great potential as a technique for analytical proteomics.

• The utility of CE is limited at the present time by the lack of commercially available, robust, and reliable CE-MS instrumentation for analytical proteomics.

Qno8: Write down the advantages of Single molecule Sequence. 5 marks

Advantages of single molecule sequencing

• Less sample preparation (no PCR)
• No amplification

• no PCR errors

• fewer contamination issues

• no GC-bias

Qno9: Explain the structure of siRNA. 2 marks

Small interfering

RNA (siRNA), sometimes known as short interfering RNA, is a class 20- 25 nucleotide-long RNA molecules that interfere with the expression of genes.

A short (usually 21-nt) double-strand of RNA (dsRNA) with 2-nt overhangs on either end, including a 5' phosphate group and a 3' hydroxy (-OH) group.

Qno10: Write down the application siRNA. 3 marks

Applications of siRNA

• Determining protein function

• Easier than preparing knockout

• Used for partial knockdowns

• Applications for cancer prevention, infections and developmental defects.

Qno11: Why 2D SDS-PAGE was not used in past? 2 marks

In past 2D SDS-PAGE was difficult to use because of;

• The relative technical difficulty of performing the IEF step.

• Getting the delicate tube gel containing the focused proteins set up to efficiently transfer the proteins in the SDS-PAGE slab gel was a technical challenge.

Qno12: Write down the advantages of Genome assembly. 3 marks

• It is unlikely that both reads (mate pairs) are coming from a repeat.

• A read in a unique portion of DNA can help find the true repeat where its mate is residing.

Qno13: How transcriptions help us? 2 marks

RNA transcription makes an efficient control point because many proteins can be made from a single mRNA molecule. Transcript processing provides an additional level of regulation for eukaryotes, and the presence of a nucleus makes this possible.

Qno14: Define Domain. 2 marks

Large polypeptide chains usually fold into two or more globular clusters known as **domains**, which often give these proteins a bi- or multilobal appearance.

Qno15: What is open frame reading? 2 marks

It is the part of gene that has a potential to code for proteins comes in triples called codons, beginning and ending with a unique translation start (ATG) and one of three stop (TAA, TAG, TGA) codons.

Qno16: Explain the types of protein. 3 marks

• Three major categories

• Fibrous proteins

• Globular proteins

• Complexes with multiple components including proteins

Qno17: Name the assemblers in Greedy graph. 3 marks

Greedy Graph based Assemblers:

• SSAKE

• VCAKE

• SHARCGS

Qno17: Describe the clone by clone short gun sequencing and whole genome sequencing. 5marks

Clone by clone shotgun sequencing:

• Also known as hierarchical shotgun sequencing or map based shotgun sequencing.

• The target DNA is first analysed by clone based physical mapping methods.

• Individual clones that together span the region of interest are selected and subjected to shotgun sequencing.

Whole Genome Shotgun Sequencing (WGS):

• Whole genome is broken down, cloned into vectors and then sequenced.

Qno18: Explain the importance of protein in pharmaceutical. 5 marks

Pharmaceutical Importance of proteins:

• Proteins as pharmaceuticals

• Proteins applications

• Whey proteins health effects

• Iron chelate Protein

• Zinc chelate Protein

• Tumor markers